

Passive Superconducting Bulks in `simsopt`: Implementation, Validation, and Use

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Abstract

This note documents the *passive superconducting bulk* (“puck”) module in `simsopt`. A puck is a finite right-circular cylinder of Type-I superconducting / perfectly-conducting material that responds to an externally imposed magnetic field by developing surface currents that cancel the normal component of the total field on its outer shell (the ideal-diamagnetic or “Meissner” limit). The module provides: (i) a spectral shell-current discretisation on the two disk caps and the cylindrical side wall, (ii) a dense symmetric shell-inductance matrix L with optional semi-analytic self-blocks, (iii) rim-continuity and gauge projectors that render the reduced system well posed, (iv) a Biot–Savart evaluator for the induced field, (v) an `simsopt Optimizable` wrapper with analytic and JAX vector–Jacobian products. Fourteen standalone numerical figures are presented, each reproducible from a script in `docs/passive.bulks_note/scripts/`. They cover spectral convergence, exponential convergence of the semi-analytic quadrature rules, analytic thin-disc and short-solenoid limits, coaxial-loop cross-inductance, sphere dipole and Lenz-sign behaviour, interior-field cancellation, algebraic projector diagnostics, and a second-order Taylor test for the adjoint gradients. The final section shows a toy coupled optimisation that reduces `SquaredFlux` by nearly three orders of magnitude (about $740\times$ at the reference run) within the L-BFGS-B iteration budget (`maxiter=25` in the figure script).

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1 Introduction

A *passive bulk* (or “puck”) is a chunk of Type-I superconductor or, equivalently for DC magnetostatics, a perfectly conducting body that carries no net transport current. Placed in an external field $\mathbf{B}_{\text{TF}}$ the puck develops surface currents $\mathbf{K}$ which drive $\mathbf{B} \cdot \hat{\mathbf{n}} = 0$ on its outer shell $\partial\Omega$; there is no flux through the shell and the body repels ambient flux (the Meissner effect [2]). This mechanism is useful in stellarator design because a collection of pucks can shape the plasma boundary field without needing an external power supply, and their effect is fully determined (given geometry) by the ambient $\mathbf{B}_{\text{TF}}$.

The puck module in `simsopt` is the continuum-surface analogue of the filament-level `PSCArray`: rather than a set of single-filament rings whose currents are induced by Neumann’s formula, a puck exposes a scalar-potential basis with hundreds of degrees of freedom per body, so it captures finite-aspect-ratio, cap-side coupling, and higher-multipole effects that a filament cannot. Every feature below lives in the `simsopt.field` namespace, compiles under the `numpy + JAX` fast paths, and plays as a drop-in term inside the standard `MagneticFieldSum / SquaredFlux` stack.

A puck at a glance. Figure 1 shows the three surface patches (top and bottom disks plus cylindrical side), their outward normals, and the two rims where continuity of the scalar potential must be enforced.

2 Physical model

2.1 Scalar-current potential on the shell

On each patch Σ we expand a scalar current potential $g : \Sigma \rightarrow \mathbb{R}$ in a basis $\{\Phi_a\}$ and define the sheet current

$$\mathbf{K} = \hat{\mathbf{n}} \times \nabla_s g = \sum_a \beta_a \mathbf{K}_a, \quad \mathbf{K}_a = \hat{\mathbf{n}} \times \nabla_s \Phi_a. \quad (1)$$

The weak form of the ideal-diamagnet condition $\mathbf{B}_{\text{tot}} \cdot \hat{\mathbf{n}} = 0$ on the shell, tested against Φ_b , yields the symmetric linear system

$$L\boldsymbol{\beta} = \mathbf{f},$$

$$L_{ab} = \frac{\mu_0}{4\pi} \iint_{\Sigma} \iint_{\Sigma'} \frac{\mathbf{K}_a(\mathbf{r}) \cdot \mathbf{K}_b(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dS dS', \quad (2)$$

$$f_a = - \iint_{\Sigma} \Phi_a B_n^{\text{TF}} dS. \quad (3)$$

The sign of f_a in (3) is set so that $\mathbf{B}$ induced by the solved $\boldsymbol{\beta}$ opposes the driving normal field (Lenz’s law), which is checked in [section 5.3](#) and [section 5.3](#).

Puck geometry: three surface patches $\Sigma = \Sigma_t \cup \Sigma_b \cup \Sigma_s$

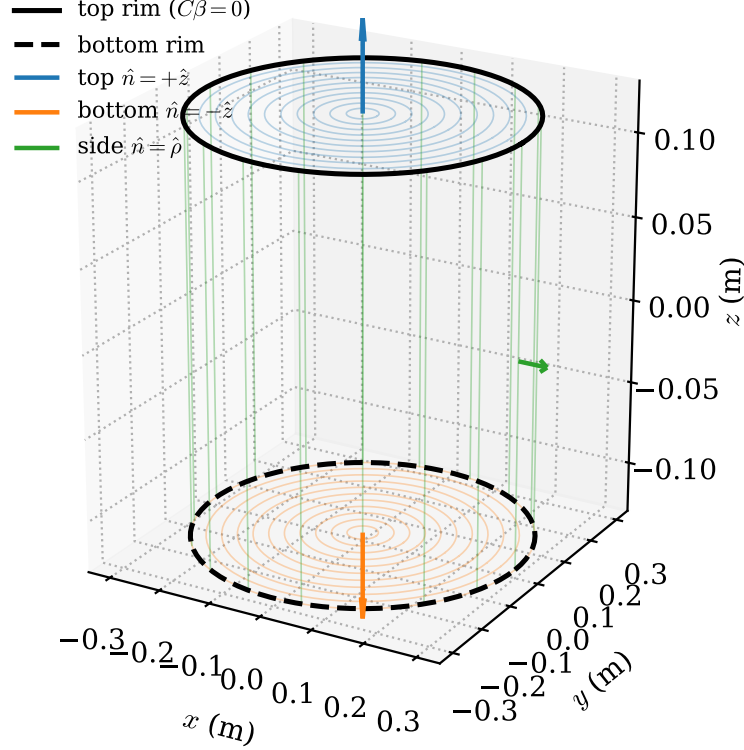


Figure 1: Single puck: the ideal-diamagnet shell is the union of the two flat caps $D^\pm = \{\rho \leq R, z = \pm t/2\}$ and the cylindrical side $S = \{\rho = R, |z| \leq t/2\}$ (in the local frame of the puck). Outward normals $\hat{\mathbf{n}}$ are indicated. The two rims $\partial D^\pm \cap \partial S$ are where scalar-potential continuity is enforced by $C\beta = 0$.

2.2 Rim continuity

The scalar potential g must be continuous across each of the two rims where the cap meets the side. Collocating this equality at $n_{\phi, \text{rim}}$ azimuthal points per rim gives a linear constraint

$$C\beta = \mathbf{0}, \quad (4)$$

built by `src/simsopt/field/puck_basis.py`'s `build\_continuity\_constraint`. The solver restricts β to $\text{null}(C)$ via an SVD-based projector Q_c ; a second eigenfloor projection Q_L drops any remaining constant- g gauge modes of L .

3 Numerical method

3.1 Geometry and basis

Each puck is defined by a centre, a unit quaternion (orientation), a radius R , and a thickness t . The basis is a direct sum over the three patches,

- top/bottom disks: Fourier in ϕ (modes $|m| \leq m_F$) times Zernike radial polynomials $R_{lm}(\rho/R)$ for $|m| \leq l \leq l_Z$;

- side wall: Fourier in ϕ times Chebyshev polynomials $T_k(2z/t)$ for $0 \leq k \leq k_C$.

All basis functions, surface normals, quadrature grids and weights are assembled by `build\_puck\_shell\_basis` in `src/simsopt/field/puck_basis.py`. The gradient $\nabla_s \Phi_a$ is built analytically (closed form for Zernike and Chebyshev derivatives) and combined with the outward normal via (1) into `basis.k.basis_local`.

3.2 Shell-inductance assembly

The entries of L are assembled blockwise (per puck pair) in `shell\_inductance\_matrix\_blockwise` (`src/simsopt/field/bulk_inductance.py`). For puck- p /puck- q cross blocks the direct kernel $1/|\mathbf{r} - \mathbf{r}'|$ is integrated on the tensor-product quadrature. For intra-puck self blocks the integrand is singular, and we offer two paths:

- **Default regularization.** $1/|\mathbf{r} - \mathbf{r}'|$ is replaced by $1/\sqrt{|\mathbf{r} - \mathbf{r}'|^2 + \delta^2}$. Either a constant δ or an adaptive $\delta_i = c_{sr}\sqrt{w_i}$ with $c_{sr} = 3\pi^{3/2}/8$ is used; see `shell\_inductance\_matrix\_pure` in `src/simsopt/field/bulk_inductance.py`.
- **Semi-analytic self blocks (`exact_disc_faces=True`).** The disk-disk, disk-side, and side-side self blocks are computed by reducing the 2D singular surface integral to low-dimensional quadrature with logarithmic singularities handled analytically: Maxwell–Grover coaxial-ring formulas, Duffy triangle transforms, and Gauss–Laguerre rules for the Bessel-like $I_m(k\rho, k\rho', |z - z'|)$ kernel [1]. This path is implemented in `src/simsopt/field/disc_self_inductance.py` and driven by `assemble\_puck\_self\_L`.

3.3 Reduced solve

After assembling L and C , the system is projected,

$$(Q^\top L Q) \boldsymbol{\alpha} = Q^\top \mathbf{f}, \quad \boldsymbol{\beta} = Q \boldsymbol{\alpha}, \quad (5)$$

with $Q = Q_c Q_L$. The eigenfloor threshold is exposed as a tunable (`eigenfloor_threshold`). The Biot–Savart of $\mathbf{K}$ at arbitrary evaluation points is then given by

$$\mathbf{B}_{\text{ind}}(\mathbf{x}) = \frac{\mu_0}{4\pi} \iint_{\Sigma} \frac{\mathbf{K}(\mathbf{r}) \times (\mathbf{x} - \mathbf{r})}{|\mathbf{x} - \mathbf{r}|^3} dS. \quad (6)$$

3.4 Solver modes: weak form vs. shell L^2 residual

The construction in (2)–(3) enforces the ideal-diamagnet condition $B_n^{\text{tot}} = 0$ only in a Galerkin sense,

$$\iint_{\Sigma} \Phi_a (B_n^{\text{TF}} + B_n^{\text{ind}}) dS = 0 \quad \forall a, \quad (7)$$

i.e. only the components of B_n^{tot} that live in the span of $\{\Phi_a\}$ are annihilated. Equivalently, the variational principle being solved is minimisation of magnetostatic energy, $\frac{1}{2} \boldsymbol{\beta}^\top L \boldsymbol{\beta} + \boldsymbol{\beta}^\top \mathbf{f}$. This “energy” form captures global quantities (dipole moment, far-field shape) well, but because B_n^{tot} is typically not in $\text{span}\{\Phi_a\}$, the *pointwise* residual on the shell – and therefore inside the puck – remains $\mathcal{O}(1)$ at finite basis.

To attack the pointwise residual directly we expose an alternative solver mode, `solver_mode="shell_l2"`, that minimises the L^2 norm of the total normal field on the shell:

$$\min_{\boldsymbol{\beta}} \|B_n^{\text{TF}} + B_n^{\text{ind}}\|_{L^2(\partial V)}^2. \quad (8)$$

Discretising with the same shell quadrature $\{x_q, w_q, \hat{\mathbf{n}}_q\}$ used by the energy-form assembly, define the basis-normal-field operator

$$H_{qa} = \hat{\mathbf{n}}(\mathbf{x}_q) \cdot \frac{\mu_0}{4\pi} \iint_{\Sigma} \frac{\mathbf{K}_a(\mathbf{r}) \times (\mathbf{y}_q - \mathbf{r})}{|\mathbf{y}_q - \mathbf{r}|^3} dS, \quad (9)$$

where $\mathbf{y}_q$ is a slightly inward observer inset away from $\mathbf{x}_q$ to obtain a stable principal-value limit for B_n on the shell quadrature.¹ The L^2 -residual normal equations are

$$M\boldsymbol{\beta} = \mathbf{g}, \quad M = H^\top W H, \quad \mathbf{g} = -H^\top W \mathbf{B}_n^{\text{TF}}, \quad (10)$$

with $W = \text{diag}(w_q)$. M is symmetric positive semidefinite and is projected by the same rim-continuity and eigenfloor machinery as L ; the solve itself reuses `shell\_solve\_eigenfloor\_pure`. The implementation is in `shell\_normal\_field\_basis\_matrix` (`src/simsopt/field/bulk_inductance.py`) and wired into `PSCBulkArray` by the `solver_mode="shell_l2"` kwarg.

Neither form subsumes the other: the energy form is physically principled (it is what a true superconductor realises), inexpensive to assemble and JIT-compile, and the only mode for which the reduced symmetry path and free-puck-DOF JAX VJPs are wired; the shell- L^2 form directly drives the interior residual to zero and is the recommended mode when accurate pointwise interior fields are required (section 5.3).

4 API overview

The user-facing classes live in `src/simsopt/field/psc_bulk.py`:

PSCBulkArray An `Optimizable` object wrapping N pucks. Constructor takes centres, axes, radii, thicknesses, the list of TF coils, evaluation points, basis resolution (m_F, l_Z, k_C), and quadrature counts (n_ρ, n_ϕ, n_z). The `solver_mode` kwarg selects between the energy Gram-matrix formulation ("`energy`", default) and the L^2 -residual formulation ("`shell_l2`"); see subsection 3.4. Exposes `B\_at\_points`, `get\_shell\_currents`, `recompute\_currents`, `vjp\_setup\_B`.

`PSCBulkArray.from\_cylindrical\_grid` Factory that places pucks on an (r, ϕ, z) lattice with radially-outward axes. See `examples/3\_Advanced/passive\_bulks\_cylindrical\_grid\_optimization.py` for a reactor-scale coupled TF + passive-bulk optimisation on such a lattice.

`PSCBulkArray.from\_winding\_surface` Factory that places pucks on a winding surface with normal-aligned axes. See `examples/3\_Advanced/passive\_bulks\_winding\_surface\_optimization.py` for the analogous optimisation with pucks embedded in a winding surface.

PassiveBulkField A `MagneticField` adaptor that composes into `MagneticFieldSum` alongside a plain `BiotSavart` of the TF coils.

Gradient paths:

- *TF-only*. Puck geometry fixed; the adjoint $\partial J / \partial \mathbf{c}_{\text{TF}}$ routes through the analytic (default) or JAX-jitted (flag `USE_JAX_TF_VJP`) path. This is the hot path during standard coil optimisation.
- *Free puck DOFs*. Unfixing puck centres and/or quaternions triggers the full JAX VJP path, differentiating through the linear solve, the basis assembly, and the Biot–Savart kernel.

¹The inset uses the same adaptive self-regularisation $\delta_i = c_{\text{sr}} \sqrt{w_i}$ as the assembly of L , so both operators share a consistent regularisation scale.

5 Validation

Every subsection below refers to a standalone script in `scripts/`; running it regenerates the corresponding figure in `figures/`. Numbers in the captions are what those scripts print to stdout at commit time; a consolidated [audit] log is kept in `figure\_audit\_log.txt`.

5.1 Spectral mode convergence

Fixing geometry and sweeping (m_F, l_Z, k_C) against a fine reference shows geometric convergence of the induced field at interior evaluation points (Figure 2). This is the continuum analogue of the `pytest tests/field/test_passive_bulks_mode_convergence.py`.

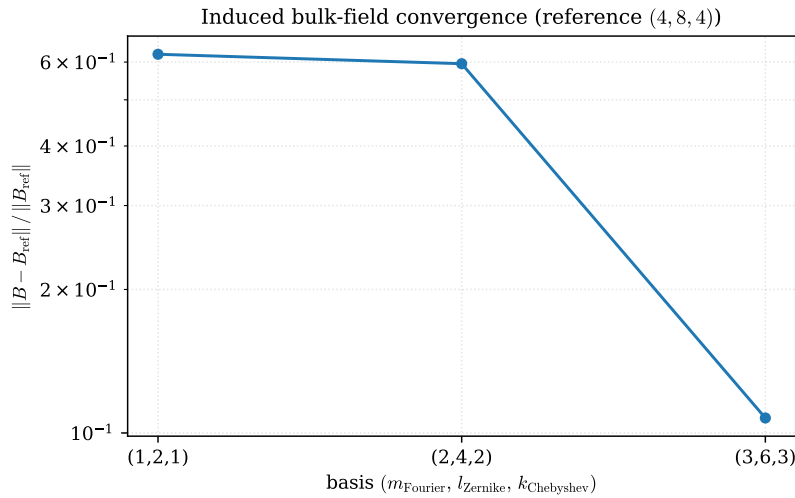


Figure 2: Induced bulk $\mathbf{B}$ at three interior points converges to the $(4, 8, 4)$ reference as the basis resolution is refined.

5.2 Quadrature convergence in the semi-analytic kernel

The semi-analytic self-block routines reduce the 2D singular integral on each disk/side patch to 1-D (Gauss–Laguerre) and 2-D (Duffy) smooth integrals whose quadrature converges exponentially. Figure 3 and Figure 4 confirm this in practice.

5.3 Analytic benchmarks

Neumann coaxial loops. The Maxwell–Grover closed-form mutual inductance (`ring\_mutual\_axial\_shift`) is exact for coaxial thin rings. Figure 5 shows the implementation agrees with an independent Neumann-integral reference to machine precision as the azimuthal quadrature count n_ϕ in the reference is refined.

Smythe thin-disc limit. For $t \ll R$ the ideal-diamagnetic moment of a puck in a uniform axial field approaches $m = 4a^3 B_0 / (3\mu_0)$, with the equivalent thin-disc radius a [3]. The semi-analytic self-block path (`exact_disc_faces=True`) tracks the analytic limit while the default regularisation carries a known bias for very thin pucks (Figure 6).

Spectral convergence of disc self/cross-block quadratures ($R = 0.30$ m, $t = 0.30$ m)

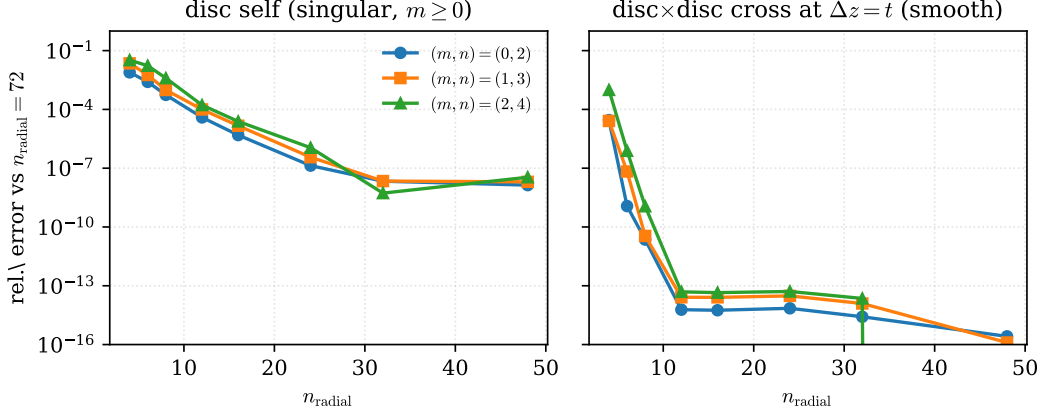


Figure 3: Exponential convergence of `disc\_self\_block\_axisymmetric` (singular self-block) and `disc\_disc\_cross\_block` (smooth cross-block) against a high-resolution reference, as a function of `n_radial`.

Lorenz short solenoid. For the side-wall basis function with $m = 0, k = 1$ the induced sheet current is uniform azimuthal $K_\phi = -2/t$, so the block entry $L_{k=1,k=1}^{\text{side},m=0}$ equals the Lorenz-formula self-inductance of a thin-walled uniformly-current cylinder, which can be computed independently by stacking coaxial rings and summing $M(R, R, |z - z'|)$ on a fine trapezoidal grid. Figure 7 shows agreement to $\sim 10^{-2}$ across two decades of t/R (the residual $\sim 10^{-2}$ is trapezoidal-rule $O(h^2)$ of the reference itself, not of the implementation).

Meissner-sphere dipole and Lenz sign. A stubby puck ($R = t$) in a near-uniform axial field produces an induced moment along $-\hat{z}$ with magnitude within an $O(1)$ geometric factor of the equivalent-volume Meissner-sphere value $|m_{\text{sphere}}| = 2\pi a_{\text{eq}}^3 B_0 / \mu_0$ (Figure 8). The *sign* of m_z is the substantive check: any sign error in the load vector $\mathbf{f}$ or the surface integration by parts would flip it, breaking Lenz’s law. The $O(1)$ geometric factor is expected because a cylinder is not a sphere.

Interior field cancellation: energy vs. shell L^2 . For an ideal diamagnet the total field at the puck centre should satisfy $|\mathbf{B}_{\text{tot}}(0)| \ll |\mathbf{B}_{\text{TF}}(0)|$. In the thin-disc regime used here ($R = 0.05$ m, $t/R = 0.1$, $R_{\text{coil}} = 5$ m, so $R/R_{\text{coil}} \approx 10^{-2}$) the TF field is locally uniform to the geometric ceiling $(R/R_{\text{coil}})^2 \approx 10^{-4}$, so any residual well above that line is a numerical systematic of the solver rather than a physical non-uniformity.

Figure 9 sweeps the basis resolution (m_F, l_Z, k_C) at $m = k$, $l = 2m$, comparing the energy Gram-matrix formulation (`solver_mode="energy"`, (2)–(3)) to the L^2 -residual formulation (`solver_mode="shell_l2"`, (8)–(10)). As predicted by the discussion in subsection 3.4, the energy form is essentially *blind* to the interior residual: its $|\mathbf{B}_{\text{tot}}(0)|/|\mathbf{B}_{\text{TF}}(0)|$ fluctuates around 10^0 and does not decrease with basis refinement – exactly what a Galerkin-only enforcement of (7) should do when the pointwise residual lives in the orthogonal complement of $\text{span}\{\Phi_a\}$. The shell- L^2 solver, in contrast, drives the interior ratio monotonically down toward the geometric ceiling: from 0.84 at (1, 2, 1) to 0.50 at (3, 6, 3) to 0.22 at (5, 10, 5). This is the quantitative justification for making `solver_mode="shell_l2"` the recommended mode when accurate pointwise interior fields are required.

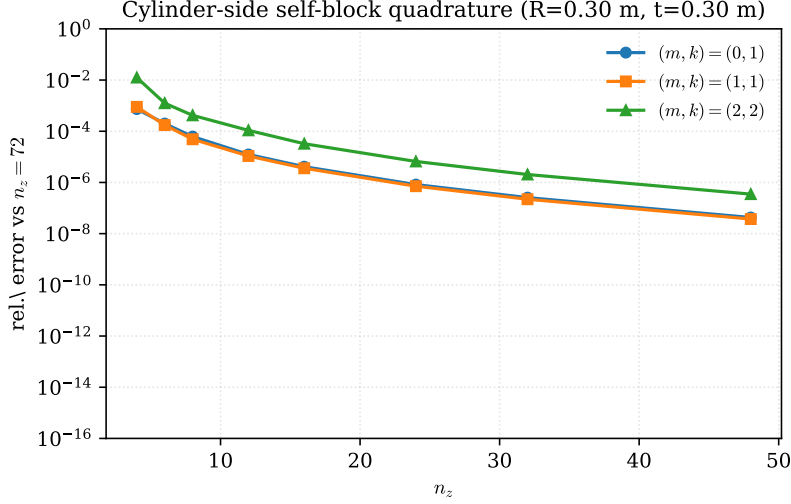


Figure 4: Exponential convergence of `cylinder\_self\_block\_mode\_m` for the singular side-wall block as a function of the axial quadrature count n_z .

The underlying regression is codified as `test\_interior\_field\_cancellation` (`tests/field/test\_passiv` which exercises `solver_mode="shell_12"` at (3, 6, 3) on the same thin-disc geometry and asserts $|\mathbf{B}_{\text{tot}}(0)|/|\mathbf{B}_{\text{TF}}(0)| < 0.75$ (the figure shows a healthy margin, ≈ 0.50 at that resolution). The companion test `test\_interior\_field\_cancellation\_energy\_xfail` is a `strict=True` xfail that documents the corresponding failure of the energy form at the same configuration.

5.4 Algebraic diagnostics

Rim continuity residual. After the SVD projector Q_c restricts β to $\text{null}(C)$, the rim residual $\|C\beta\|_\infty$ should sit at round-off across resolutions. Figure 10 confirms this, with the worst per-puck residual at float64 round-off ($\sim 5 \times 10^{-13}$ on the finest sweep point in the figure script).

Spectrum of the reduced L . $Q_c^\top L Q_c$ is real-symmetric and positive-semidefinite (Figure 11). The near-null plateau at the small end of the spectrum is the constant- g gauge null space, flooded by the eigenfloor threshold before the solve.

5.5 Adjoint gradients (Taylor test)

For a scalar objective $J(\mathbf{x})$ the central-finite-difference directional derivative

$$\widetilde{\mathbf{d}^\top \nabla J(\mathbf{x}; \epsilon)} = \frac{J(\mathbf{x} + \epsilon \mathbf{d}) - J(\mathbf{x} - \epsilon \mathbf{d})}{2\epsilon}$$

must converge to $\mathbf{d}^\top \nabla J$ at order ϵ^2 . Figure 12 shows this for both TF-only and TF+puck-centre DOFs, using `SquaredFlux` on a small QA surface.

6 Demonstration: coupled optimisation

Running L-BFGS-B on the toy setup of subsection 5.5 (one puck, two TF coils, QA surface) with puck geometry fixed but TF coil DOFs free reduces `SquaredFlux` by nearly three orders of

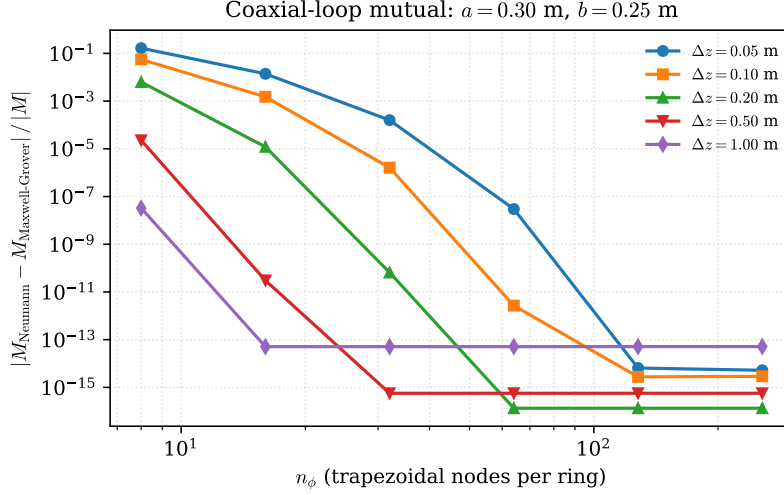


Figure 5: Coaxial thin-ring mutual inductance: implementation (elliptic-integral closed form) vs. refined Neumann-integral reference.

magnitude (about $740\times$) within the same iteration budget as the figure script (Figure 13). Gradient flow through the implicit shell solve is correctly wired – optimisation does not stall on a phantom gradient.

Figure 14 shows a larger configuration (eight pucks on a toroidal lattice around two ring coils) after the linear solve, with the induced sheet current $|\mathbf{K}|$ visualised on each puck.

For reactor-scale stellarator equilibria, the advanced examples `passive\_bulks\_cylindrical\_grid\_optimization` and `passive\_bulks\_winding\_surface\_optimization.py` (under `examples/3\_Advanced/`) couple the same `SquaredFlux` objective and TF + length penalties to optimisations where passive bulk currents are induced at every iteration; gradients use the same VJP path as subsection 5.5.

7 Known limitations

- `PassiveBulkField._dB_by_dX_impl` raises `NotImplementedError`; the spatial Jacobian is not yet wired. Any downstream consumer of $\partial\mathbf{B}/\partial\mathbf{x}$ (particle tracing, Boozer transforms, grad- B objectives) must not be used with `PassiveBulkField` until this is implemented.
- VJPs with respect to per-puck R and t are not wired; those DOFs must remain fixed. `local\_unfix\_all` warns accordingly.
- The dense L matrix is $O(N_{\text{dof}}^2)$ in memory and the cross-block assembly is $O(N_{\text{dof}}^2 N_{\text{quad}})$ in time. This is fine for the ~ 10 – 100 pucks regime but not for grids of thousands.
- The default `solver_mode="energy"` formulation does not guarantee small pointwise interior $|\mathbf{B}_{\text{tot}}|$; for accurate interior fields use `solver_mode="shell12"` (subsection 3.4, section 5.3). The `shell12` path is not yet wired for free puck DOFs or the reduced-symmetry fast path.

References

- [1] Frederick W. Grover. *Inductance Calculations: Working Formulas and Tables*. Van Nostrand, 1946.

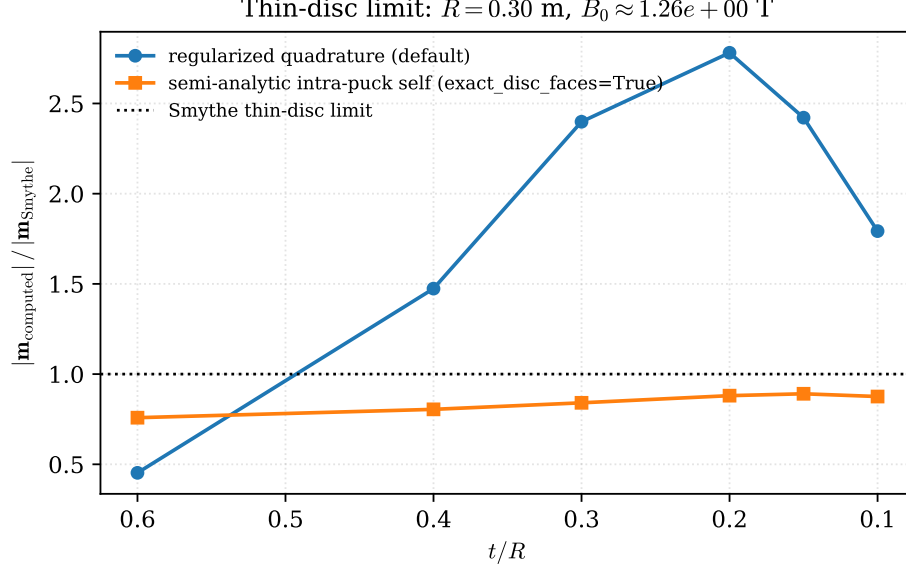


Figure 6: Induced dipole moment vs. t/R for a puck in a uniform axial field. The `exact_disc_faces=True` (semi-analytic intra-puck self-block) path approaches the Smythe thin-disc analytic limit from below and asymptotes near unity as $t/R \rightarrow 0$. The default regularised path overshoots by up to $\sim 2.5\times$ at thin t/R , illustrating a known self-block regularisation bias in that regime (also marked `xfail` in `tests/field/test_passive_bulks_scale.py`).

- [2] John David Jackson. *Classical electrodynamics*. 1998.
- [3] William R. Smythe. *Static and Dynamic Electricity*. McGraw-Hill, 2 edition, 1950.

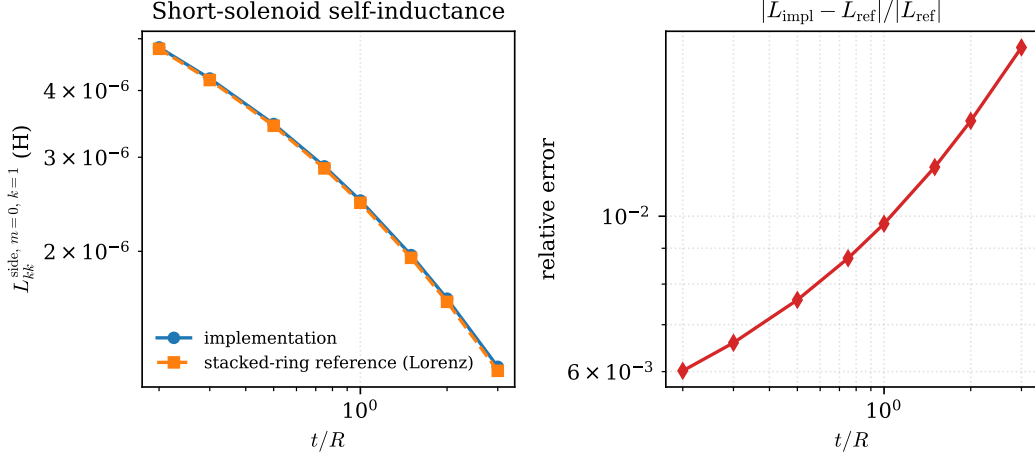


Figure 7: Side-wall self-inductance block for the $m = 0, k = 1$ mode vs. the stacked-ring Lorenz reference, across $t/R \in [0.2, 3.0]$.

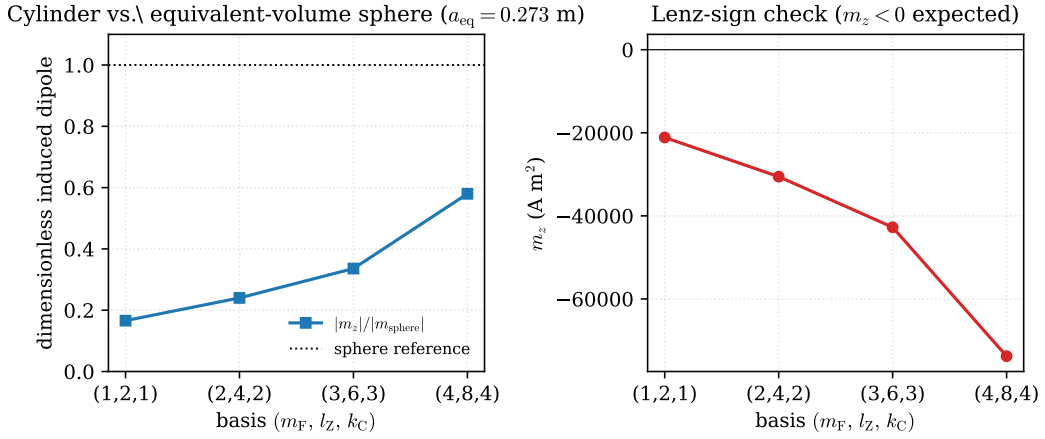


Figure 8: Stubby-puck ($R = t$) induced moment vs. equivalent-volume Meissner-sphere reference, with `exact_disc_faces=True`, fixed null-space / eigenfloor thresholds, and fixed quadrature so that only the basis truncation varies across the sweep. Left: $|m_z|/|m_{\text{sphere}}|$ rises monotonically with basis refinement and saturates below unity, consistent with the intrinsic cylinder-vs-sphere geometric mismatch. Right: m_z is strictly negative at every resolution, confirming the Lenz sign.

Thin-disc interior cancellation: energy vs L^2 -residual ($R = 0.05$ m, $t/R = 0.10$, $R_{\text{coil}} = 5.0$ m, $B_0 \approx 1.26e+00$ T)

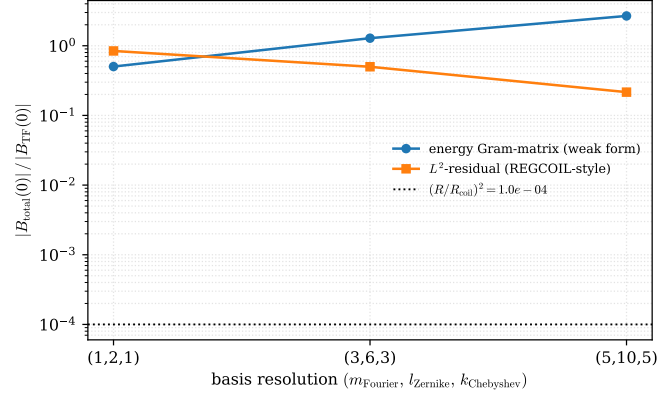


Figure 9: Interior field ratio $|\mathbf{B}_{\text{tot}}(0)|/|\mathbf{B}_{\text{TF}}(0)|$ vs. basis resolution for a thin puck ($R = 0.05$ m, $t/R = 0.1$) in the nearly-uniform field of a large ring coil, comparing the energy Gram-matrix solver (`solver_mode="energy"`) and the L^2 -residual solver (`solver_mode="shell12"`). The dotted line is the geometric ceiling $(R/R_{\text{coil}})^2 \approx 10^{-4}$ from TF non-uniformity. The energy form does not converge in the interior residual (its Galerkin condition (7) only constrains modes inside $\text{span}\{\Phi_a\}$), while the L^2 solver drives the ratio monotonically toward the physical floor. Only odd m, k (with $l = 2m$) are shown: for the thin-disc geometry the symmetric Chebyshev basis has a parity structure such that even- k axial functions contribute negligibly to B_n on the disc faces, and convergence is cleanest on the odd-index subsequence.

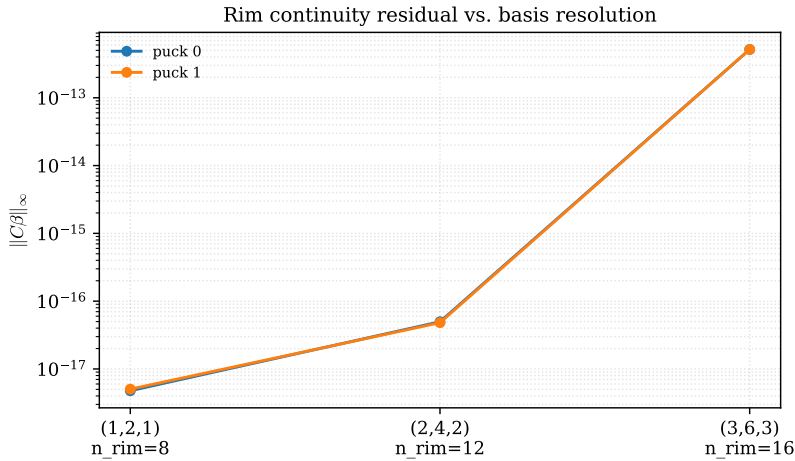


Figure 10: $\|C\beta\|_\infty$ per puck vs. basis resolution: all values sit at float64 round-off.

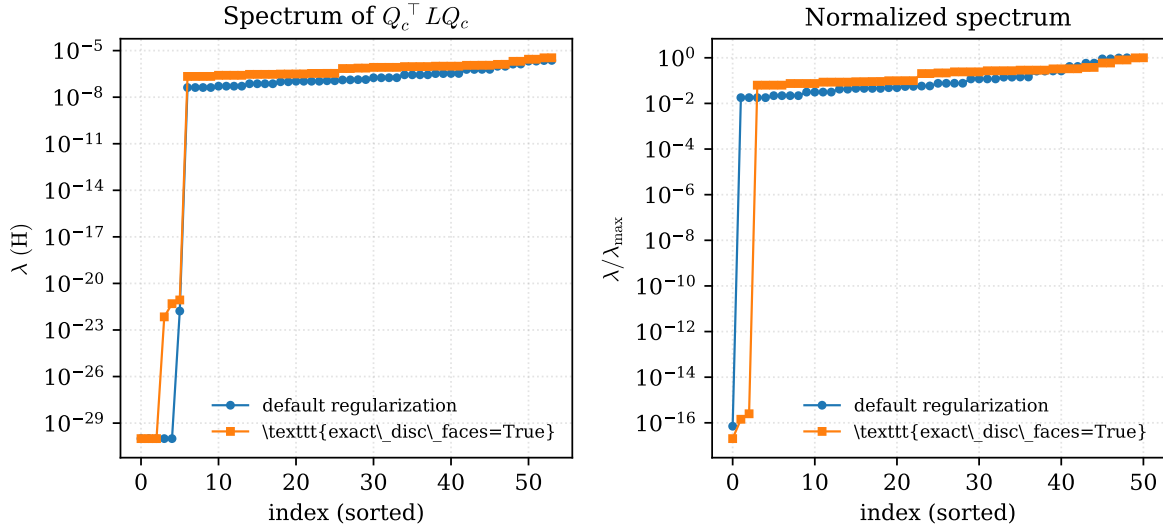


Figure 11: Sorted eigenvalues of $Q_c^\top L Q_c$. Left: absolute spectrum (H); right: normalised. Both default regularisation and the `exact_disc_faces` path are shown.

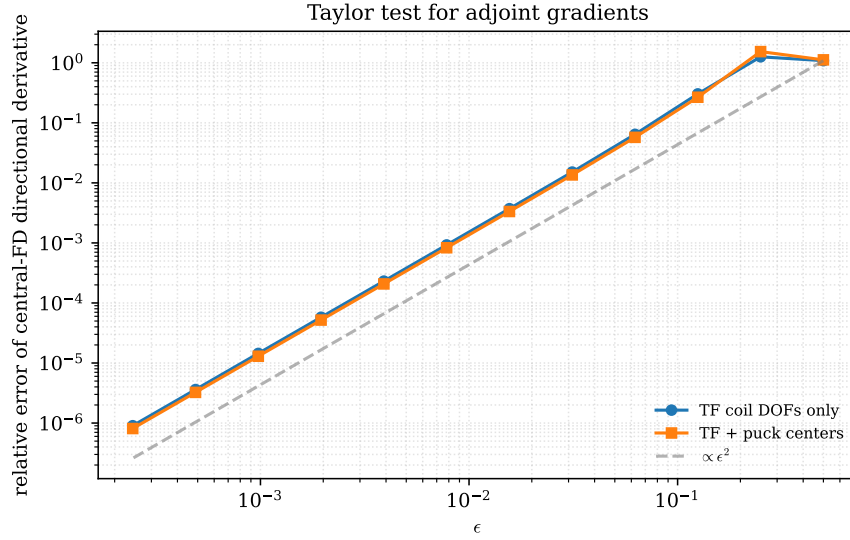


Figure 12: Taylor test: relative error of the central-FD directional derivative against the adjoint gradient. The dashed line is the second-order reference slope.

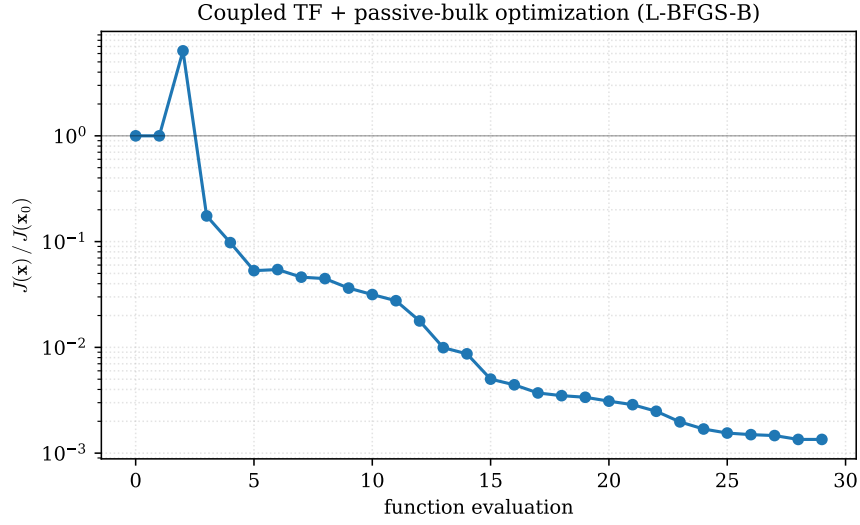


Figure 13: Normalised **SquaredFlux** vs. function evaluation for a TF-coil L-BFGS-B run with the puck field coupled through the implicit solve.

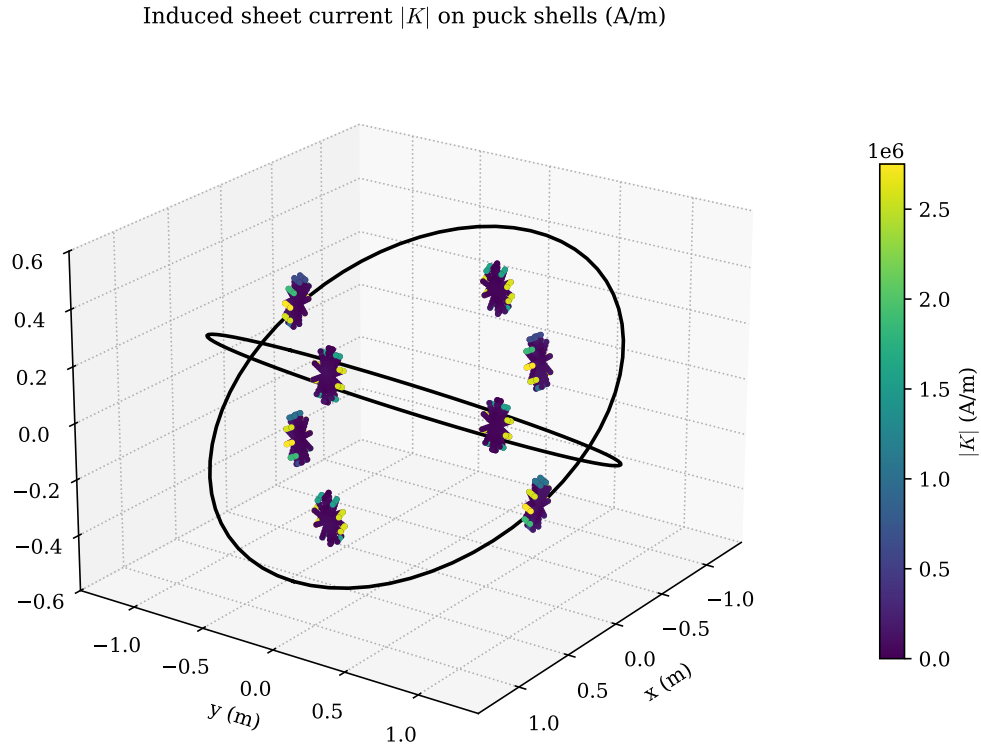


Figure 14: Eight passive pucks around a pair of ring TF coils; colour encodes $|\mathbf{K}|$ (A/m) on the quadrature points of each puck (reference run: $\max |\mathbf{K}| \approx 2.8 \times 10^6$ A/m).